Solano County Health & Social Services Department

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POLICY MEMORANDUM 6200

DATE: June 30, 2006

REVIEWED & APPROVED BY:

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SUBJECT: AMBULANCE RESPONSE TO HAZARDOUS MATERIALS SPILLS

AUTHORITY: CODE OF FEDERAL REGULATIONS, PART 3, DEPT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION; CALIF. HEALTH & SAFETY CODE, DIVISION 2.5, 1797.220

PURPOSE/POLICY:
To establish guidelines to manage the prehospital scene of HAZ-MAT spills.

I. INTRODUCTION

Be alert for hazardous materials when responding to every call. Hazardous materials can be obvious (i.e., noxious fumes, gasoline or corrosive liquid spills), or they can be hidden (odorless but poisonous and/or flammable vapors and liquids, radiation). If a vehicle has a diamond shaped placard or an orange numbered panel on the side or rear, the cargo **should be assumed to be hazardous**. Assume the cargo is leaking and take immediate precautions. Unfortunately, not all hazardous materials will be that clearly marked. For example, grocery trucks and United Parcel trucks regularly carry hazardous materials that can be released in a collision. Judgment should be used when weighing the risks of delay in the care of a patient with serious injury against the degree of risk of hazardous materials contamination of yourself or others., i.e., a situation where hazardous materials exposure to yourself can be controlled by wearing gloves versus a situation where exposure of any type is determined to be extremely dangerous.

II. INITIAL AMBULANCE RESPONSE

A. Unless otherwise directed, park upwind and uphill from any incident where you suspect hazardous materials.

B. Do not drive or walk through any spilled material.

C. If first-in responder, confirm that fire and police have been notified and are aware that hazardous materials might be involved.

D. If first-in responder, first priority is scene isolation. **KEEP OTHERS AWAY! KEEP UNNECESSARY EQUIPMENT FROM BEING CONTAMINATED.**

III. APPROACH AND TREATMENT OF VICTIMS

A. Do not approach any victims without permission of Incident Commander.

B. Do not approach anyone coming from contaminated areas (particularly those potentially contaminated) until given permission by the Incident Commander (vapors can be trapped in clothing and carried out of contaminated area!).

C. Follow Incident Commander's instructions regarding victim decontamination. Fire department is responsible for initial decontamination.

1. Insure all potentially contaminated patient clothing and belongings have been removed. Do not transport clothes and belongings along with patient unless Incident Commander directs otherwise.
2. Insure irritated skin has been flushed with water for a minimum of 15 minutes. Insure irritated eyes are rinsed for a minimum of 15 minutes. Use saline solution if available.
3. **DECONTAMINATION TAKES PRIORITY OVER TREATMENT AND/OR TRANSPORT.**

D. Avoid contact with contaminants. Wear disposable gloves as minimum protection and use protective clothing as appropriate.

E. Utilize "kerlex" type bandages rather than adhesive bandages in treating wounds.

F. Provide oxygen by mask for any victim with respiratory problems.

G. Continue to rinse irritated eyes with water while enroute to hospital and be alert for respiratory distress. USE BASINS TO SAVE THE WATER USED FOR RINSING.

H. Cover entire gurney, including pillow, with a disposable (plastic lined) blanket.

I. Cover patient compartment floor with a disposable (plastic lined) blanket. Do this either before initially arriving or before you re-enter vehicle in order to minimize contamination from shoes. Not necessary in all cases (e.g., gas exposure).

J. Write down the name of the involved chemicals, if identified, before leaving the scene.

K. If source is available, ask for advice on decontaminating vehicle and personnel once patient is released to the hospital (Environmental Health on-scene representative, Incident Commander's representative, etc).

L. Provide maximum fresh air ventilation to patient and driver's compartment regardless of the presence or absence of odors.

M. Specific ALS Treatments for common hazardous materials are detailed in Attachment A. Use the appropriate Protocol as indicated.

N. Unless Multi-Casualty Plan communications are in effect, follow normal radio procedures with receiving hospital but in addition give the name of the chemical involved.

IV. ARRIVAL AT HOSPITAL

A. Patient attendant would remain in vehicle with patient until driver has personally advised emergency room personnel of the situation and has handed them the written identification of the chemical involved.

B. The patient should not be brought into the emergency room before the attendants receive permission from the emergency room staff.

C. Once the patient has been released to the hospital, double bag (in plastic) the blankets off the floor and gurney. Double bag equipment you believe contaminated.

D. Decontaminate ambulance and personnel before returning to incident scene. Agencies or fire departments can receive advice on decontamination from the County Environmental Health Officer.

E. If not returning to scene, keep contaminated articles sealed until given further instructions by the Incident Commander or your agency. Do not go back into service without rigorous decontamination of vehicle.

HAZARDOUS MATERIALS FLOW CHART

IS POLICE SECURITY LINE ESTABLISHED?

YES

Ask police for directions to staging area or Incident Commander

NO

Secure Scene

1. Establish perimeter
2. Call for Assistance
3. Do not approach contaminated victims.

Report to Incident Commander or Staging Area outside of contaminated area

TAKE CONTAMINATION PRECAUTIONS

1. Disposable blankets over gurney and floor.
2. Wear personal protective gear if appropriate.
3. Insure patients' clothes and belongings have been removed.

TREAT PATIENTS SYMPTOMATICALLY

1. Insure irritated skin and eyes are flushed/rinsed with water for 15 minutes, minimum.
2. Provide Oxygen by mask for respiratory complaints.
3. Use "kerlex" type bandaging rather than adhesives.

TRANSPORT

1. Write name of chemical, and, if source available, ask for advice on decontaminating vehicle.
2. Provide maximum fresh air ventilation.
3. Continue symptomatic treatment.

HOSPITAL ARRIVAL

1. Notify E. R. Staff by radio and request instructions.
2. Provide name of chemical to E.R. Staff.
3. Double bag contaminated blankets, sheets and protective gear.
4. Implement decontamination advice.
5. Do not return to non-incident related service until given permission by your ambulance agency.

IRRITANT GASES

ACIDS & ACID MISTS, AMMONIA AND CHLORINE GAS

Notes on Acids & Acid Mists (NOT including Hydrofluoric Acid)

BACKGROUND:
Acids act as direct irritants and corrosive agents to moist mucous membranes and to intact skin to a lesser extent. Concentrated acids may cause sever skin burns. Generally, these substances have very good warning properties: even fairly low airborne concentrations of acids produce rapid onset of eye, nose and throat irritation. Higher concentrations can produce cough, shortness of breath, wheezing, chemical pneumonia or non-cardiogenic pulmonary edema. Occasionally, pulmonary edema may be delayed for several hours, especially with low-solubility gases such as nitrogen oxides produced from nitric acid. Ingestion of acids can result in severe injury to the upper airway, esophagus and stomach.

Notes on Ammonia (Liquid and Gas)

BACKGROUND:
Ammonia (NH3) is a direct irritant and alkaline corrosive agent to moist mucous membranes and, to a lesser extent, to intact skin. Ammonia has very good warning properties. Even fairly low airborne concentrations produce rapid onset of eye, nose and throat irritation. Higher concentrations can produce cough, shortness of breath, wheezing, chemical pneumonia or non-cardiogenic pulmonary edema. The onset of pulmonary edema is usually rapid but may occasionally be delayed for 12-24 hours. Ingestion of concentrated ammonia solutions (e.g., >5%) may cause serious corrosive injury to the esophagus and stomach.

Notes on Chlorine Gas

BACKGROUND
Chlorine is a highly irritating gas which rapidly forms hydrochloric acid after contact with the moist mucous membranes in the upper airway and in the lungs. Symptoms occur rapidly and provide good warning properties for exposure. Low concentrations produce eye, nose and throat irritation. Higher concentrations produce cough, shortness of breath, wheezing, choking, chemical pneumonitis, or pulmonary edema. Ingestion of concentrated hypochlorite (bleach) solutions can cause serious corrosive injury to the esophagus and stomach.

INITIAL DECONTAMINATION PRIOR TO PREHOSPITAL MANAGEMENT:

Decontamination when necessary should include flushing the victim with water spray, and if clothing has been soaked or gas is likely to be trapped in clothing, the clothing should be removed, double-bagged and skin flushed for 1 – 2 minutes. With Acids, Acid Mists and Ammonia, injured eyes should be irrigated.

POTENTIAL FOR SECONDARY CONTAMINATION:
Small amounts of acid mists, ammonia vapor or chlorine gas can be trapped in clothing after an overwhelming exposure but are usually not sufficient to create a hazard for health care personnel

away from the scene. However, skin or clothing which has become exposed to the liquid may be corrosive to rescuers. Once the victim has been stripped and flushed with water, there is no significant risk of secondary contamination. Decontamination is not necessary for victims with inhalation of acid mists and chlorine gas exposure only.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION:

ACIDS & ACID MISTS
(NOT including Hydrofluoric Acid)
FORMS: Gas, Liquid (Variable Concentrations) mixtures with water and aerosolized dusts.

AMMONIA
FORMS: Gas (anhydrous) and liquid (aqueous solutions, variable concentrations). NOTE: liquefied compressed gas may produce cryogenic (freezing) hazard as it is released into the atmosphere.

CHLORINE GAS
FORMS: Gas (anhydrous) or liquid (aqueous chlorine usually in the form of hypochlorite, the liquid hypochlorite (bleach) solutions are very unstable and react with acids to release chlorine gas. NOTE: liquefied compressed gas may produce cryogenic (freezing) hazard as it is released into the atmosphere.

EVALUATE AIRWAY *

- Oxygen – High flow/NRM
- Flush affected skin and eyes with copious water or saline
- Cardiac Monitor
- Transport

BASE

- Consider: I.V. TKO
- Albuterol 0.3 cc in 2.5 cc NaCl

Ingestion: DO NOT induce vomiting. Immediately dilute with 1 glass of water or milk in patients who are awake and have intact gag reflex.

* Intubation should be considered if the victim develops severe respiratory distress.

Ingestion of acids, concentrated ammonia solutions and concentrated hypochlorite solutions can cause serious corrosive injury to the esophagus and stomach; therefore, an EOA should not be considered as an airway adjunct.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION

HYDROFLUORIC ACID

FORMS: Gas, liquid (variable concentrations). Fluoride salts in the presence of some acids may generate toxic quantities of hydrogen sulfide gas.

- Evaluate Airway *
- Oxygen – High Flow/NRM
- Irrigate injured eyes and skin
- Cardiac Monitor
- I.V.
- Transport
- BASE:

  Consider:

  - Calcium chloride 5cc 10% solution IV for hypocalcemic tetany or cardiac arrest or prophylactic for victims with high concentration (10-20%) exposure to >3 – 5% body surface area.
  - Specific soaking solution or fluoride binding agent. **

* Intubation should be considered if the victim develops severe respiratory distress

Ingestion can cause severe corrosive burns of the esophagus and stomach; therefore, an EOA should not be considered as an airway adjunct.

** After initial strip and flush and basic decontamination, solutions of Epsom salt (Magnesium sulfate) or iced Hyamin are effective soaking solutions, if available. Topical application of magnesium or calcium containing antacids or gels may also be considered, if available. Some industries may have these substances available for emergency treatment, should this treatment be initiated by the company, it may be continued enroute.

CARBON MONOXIDE AND SMOKE INHALATION

NOTES ON CARBON MONOXIDE

BACKGROUND:

Carbon monoxide (CO) is a colorless, odorless gas. It is a common product of combustion of any organic material and is a major toxic component in cases of smoke inhalation. Carbon monoxide causes poisoning by interfering with the binding of oxygen to hemoglobin in the blood, myoglobin in heart and muscle tissue and possibly by interfering with oxygen utilization in the cell. Symptoms of progressively worse exposure include, in order: headache, dizziness, giddiness, tinnitus, nausea, muscle weakness, chest pain, dyspnea, syncope, seizures and coma. Cherry-red skin coloration is not commonly seen (except post-mortem) and should not be relied upon for diagnosis. The half-life of CO in the blood is from 5 to 9 hours when the victim is breathing room air, compared to 60 – 90 minutes when breathing 100% oxygen.

POTENTIAL FOR SECONDARY CONTAMINATION:

Very small amounts of CO can be trapped in a victim's clothing after an overwhelming exposure, but are not usually sufficient to create a hazard for health care personnel away from the scene. Decontamination is not necessary for CO exposure alone.

NOTES ON SMOKE INHALATION

BACKGROUND:

Smoke inhalation injury results from the combined effects of the various inhaled components as well as direct thermal injury to the airway. Common toxic components of smoke include carbon monoxide, hydrogen cyanide, irritant gases such as acrolein, ammonia, hydrogen chloride, phosgene, nitrogen oxides, metal fumes and carbonaceious particulates. The proportion of each component depends on the substances and conditions of the combustion. Fires involving chemicals may produce smoke with very high concentrations of toxic substances. Stridor, wheezing, tracheobronchitis and acute upper airway obstruction may occur from irritant gases and thermal effects. Acute non-cardiogenic pulmonary edema and chemical pneumonia may occur shortly after exposure or may be delayed up to several hours, especially after exposure to nitrogen oxides.

In any victim with altered mental status, suspect inhalation of a systemic toxin such as carbon monoxide, hydrogen cyanide (see protocol), agents producing methamoglobinemia (see protocol), or smoke containing with a stored chemical such as pesticides or those found in combustion of drug laboratory chemicals.

POTENTIAL FOR SECONDARY CONTAMINATION:

Although small quantities of acid, amine or ammonia mists may be absorbed onto clothing from smoke, they pose no significant risk of secondary contamination of rescuers outside of the hot zone, even though odor and carbon deposits may be a nuisance. If the victim has been exposed to heavy concentrations of liquid or vapor from a chemical, then removal of clothing and flushing with water is appropriate.

INITIAL DECONTAMINATION PRIOR TO PREHOSPITAL MANAGEMENT:

Decontamination with smoke inhalation should include removal of contaminated clothing and flushing skin for 1 – 2 minutes if the victim has been exposed to a heavy concentration of liquid or vapor from a chemical.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION

SMOKE INHALATION

FORMS: Toxic Gases, vapors and particulates formed by combustion.

CARBON MONOXIDE

FORMS: Gas

- Evaluate Airway *
- Oxygen – High Flow/Tight Fitting NRM
- Cardiac Monitor
- Transport

BASE:

CONSIDER:
Draw Blood – lavender top for carboxyhemoglobin
IV
Albuterol 0.3 cc in 2.5 cc NaCl for smoke inhalation patients with wheezing.

If hypotensive:
FLUID CHALLENGE, if no response Dopamine 2 – 10 mcg/kg/min for HYPOTENSION with Carbon Monoxide

* Intubation should be considered if the victim develops sever respiratory distress.

ARSINE AND PHOSPHINE GAS

NOTES ON ARSINE GAS

BACKGROUND:
Arsine (AsH3) is an extremely toxic and nearly odorless gas (it has a slight odor of garlic). It is used widely in the nicroelectronics industry and occasionally occurs as a by-product in metallurgy and pesticide manufacturing. Arsine's effects are quite distinct from other arsenic compounds. Even in very small quantities, inhaled arsine produces acute hemolysis (rupture of red blood cells), which can result in cardiac decompensation due to anemia, or renal failure due to massive kidney deposition of hemoglobin. Symptoms may be delayed for 2 – 24 hours, and include weakness, abdominal and flank pain, brown urine and jaundice. Massive acute exposure appears capable of causing immediate death by an unknown mechanism.

NOTES ON PHOSPHINE

BACKGROUND:
Phosphine (PH3) is an extremely toxic gas with a nauseating odor, used in the electronics industry as an insect fumigant, and occasionally occurring as a by-product in manufacturing. Its toxicology is not well understood, but it appears to affect the central nervous system, the heart, lungs and liver. Symptoms flowing low to moderate exposure include nausea, vomiting, headache, cough, dizziness, diarrhea, muscle aches, fever and chills. Severe exposure may produce shncope, stupor, coma, pulmonary edema and death. Unlike arsine, phosphine does not produce hemolysis.

INITIAL DECONTAMINATIO PROR TO PREHOSPITAL MANAGEMENT:
Decontamination should include flushing of the victim with water spray, and if gas is likely to be trapped in clothing, clothes should be removed and double-bagged.

POTENTIAL FOR SECONDARY CONTAMINATION:
Very small amounts of arsine or phosphine can be trapped in a victim's clothing after an overwhelming exposure, but are not usually sufficient to create a hazard for health care personnel away from the scene. Therefore, decontamination of the victim is generally not required.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION

ARSINE

FORMS: Gas, may be generated in metal ore processing and electronic component manufacturing.

PHOSPHINE

FORMS: Gas. Extremely flammable, may ignite spontaneously in air or explode on contact with flame.

- Evaluate Airway *
- Oxygen – High Flow/ NRM
- Cardiac Monitor
- Transport

BASE:

CONSIDER:

- IV
- I.V. NS or LR Bolus 500cc for ARSINE exposure with severe hemolysis.

* Intubation should be considered if the victim develops sever respiratory distress.

NOTE: With ARSINE GAS exposures, massive hemolysis may occur causing the urine to appear dark orange, red or brown. If urine sample is available, the color should be noted.

CYANIDE

NOTES ON CYANIDE

BACKGROUND:
Cyanide (CN) is an extremely toxic compound which is widely used in industry in a variety of forms (gas, liquid, solid). Cyanide gas (HCN) is a major toxic component in cases of smoke inhalation. CN produces toxicity by interfering with cellular oxygen utilization. Symptoms and signs include headache, dizziness, vomiting, tachypnea, tachycardia, and coma. There may be a distinctive odor ("bitter almonds") on the victim's clothing or breath. Death can occur within minutes of exposure. If exposure is by inhalation of CN gas, peak toxic effects are seen within minutes, but after ingestion of a CN salt or nitriles and other compounds that can be metabolized to form CN, the onset of symptoms may be delayed for several hours.

INITIAL DECONTAMINATION PRIOR TO PREHOSPITAL MANAGEMENT:
Decontamination should include flushing of the victim with water spray, and if gas is likely to be trapped in clothing, clothes should be removed and double bagged.

POTENTIAL FOR SECONDARY CONTAMINATION:
If the exposure was by inhalation of HCN gas, even though there may be small amounts of gas trapped in clothing after an overwhelming exposure, this is not usually sufficient to create a hazard for health care personnel away from the scene. The risk of secondary contamination to rescuers is greater if there is liquid or solid material on the victim's clothing or skin. Victims who have ingested cyanide salts or nitriles may vomit toxic material. This vomitus may off-gas HCN gas produced by the action of stomach acid on the material. Be prepared to contain the vomitus.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION:

CYANIDE

FORMS: Colorless to pale yellow gas (hydrogen cyanide), liquid (solutions of cyanide salts), and solid (cyanide salts). Hydrogen cyanide gas may be formed when acid is added to a cyanide salt or a nitrile.

- Evaluate Airway *
- Oxygen – High Flow/NRM
- Cardiac Monitor
- IV tko
- Transport

BASE

* Intubation should be considered if the victim develops severe respiratory distress.

HYDROGEN SULFIDE, SULFIDES & MERCAPTANS

NOTES ON HYDROGEN SULFIDE, SULFIDES & MERCAPTANS

BACKGROUND:
Hydrogen sulfide (H_2S) is a highly toxic gas with an odor of rotten eggs at low concentrations. At higher concentrations olfactory fatigue rapidly occurs, making odor a poor warning symptom of danger. Hydrogen sulfide may be found in oil refineries, natural gas drilling sits, sewage sludge, liquid manure and in a variety of industry. Mercaptans are sulfur-containing, highly odorous compounds. All of these compounds are direct irritants, but the major toxicity is due to interference with cellular oxygen utilization. Low-level exposures produce irritation of the eyes, nose and throat, cough, headache, nausea and dizziness. Higher exposures can cause syncope, seizures, coma, tracheobronchitis and pulmonary edema (which may occur up to 48-72 hours later). Death may occur within minutes of acute massive exposure.

INITIAL DECONTAMINATION PRIOR TO PREHOSPITAL MANAGEMENT:
Decontamination should include flushing the victim with water spray and, if clothing has been soaked, clothing should be removed and double-bagged and flushing skin for 1 – 2 minutes. Injured eyes should be irrigated.

POTENTIAL FOR SECONDARY CONTAMINATION:
Small amounts of H_2S gas can be trapped in clothing after an overwhelming exposure but are not usually sufficient to create a hazard for health care personnel away from the scene. However, clothing which has become soaked with concentrated liquid sulfide solutions or mercaptans may pose a risk to rescuers. Once the victim has been stripped and flushed with water, there is no significant risk of secondary contamination.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION:

HYDROGEN SULFIDE & MERCAPTANS

FORMS: Gas (hydrogen sulfide, methyl and short-chain alkyl mercaptans) and liquid (other mercaptans).

- Evaluate Airway *
- Oxygen – High flow/NRM
- Irrigate injured eyes
- Cardiac Monitor
- Transport

<u>BASE</u>:
Consider: IV tko

* Intubation should be considered if the victim develops severe respiratory distress.

HYDROCARBONS

NOTES ON PETROLEUM DISTILLATES (COMMONLY USED) AND RELATED HYDROCARBON PRODUCTS (INCLUDING GASOLINE, KEROSENE, NAPTHA, AND MINERAL SPIRITS).

BACKGROUND:
The term "petroleum distillates" refers to a variety of liquid hydrocarbon mixtures derived by fractional distillation or by catalytic cracking of crude oil. They are plant-derived products (terpines) such as citrus, pine and eucalyptus oils are also included in this group.

Inhalation of vapors may produce eye, nose and throat irritation, headache, dizziness, giddiness, disorientation, altered mental status, nausea and weakness. Intense exposure such as in an enclosed space may result in respiratory depression, unconsciousness, convulsions and liver or kidney damage. Prolonged skin exposure may cause rash, dryness and skin burns. Because most are lipid soluble, they may be absorbed through the skin. Some petroleum distillates (e.g., benzene) are known or suspected carcinogens.

Aspiration into the lungs may occur during ingestion. Pulmonary aspiration of even small amounts can result in severe chemical pneumonia. These patients usually have coughing, tachypnea and other symptoms of respiratory distress.

NOTES ON HALOGENATED HYDROCARBON SOLVENTS (INCLUDING CHLORINATED SOLVENTS, DEGREASERS, PAINT STRIPPERS AND CHLOROFLUOROCARBONS).

BACKGROUND
These substances are commonly used in industry for cleaning and degreasing electronic parts or metal or other surfaces, for dry cleaning, and for refrigeration. Among the halogenated hydrocarbon solvents are trichloroethylene, perchloroethylene ('perc"), methylene chloride (dichlormethane), 1,1,1-trichloromethane (methylchloroform), Freons, Halons and other chlorofluorocarbons. Odor varies by compound and in general is not a good warning property.

These solvents are well absorbed through skin or the lungs. They tend to be excreted rapidly, largely through exhalation, usually within a period of 15 minutes to a few hours. Symptoms of overexposure are similar to those listed for petroleum distillates above. In addition, sensitization of the myocardium may result in cardiac arrhythmias. A significant concern for many of these agents is their high vapor pressure which, in confined or poorly ventilated spaces, can displace oxygen resulting in life threatening hypoxemia. Methylene chloride (often present in paint removers) deserves special attention because it is metabolized to carbon monoxide and is corrosive to skin and mucous membranes.

INITIAL DECONTAMINATION PRIOR TO PREHOSPITAL MANAGEMENT:
Decontamination should include flushing the victim with water spray, clothes should be removed and double-bagged and skin flushed for 1 – 2 minutes. Injured eyes should be irrigated.

POTENTIAL FOR SECONDARY CONTAMINATION:
If the victim was only exposed to hydrocarbon or solvent vapors, there is no risk of secondary contamination of downstream personnel. If the victim's clothing, skin or hair is soaked with solvent, rescuers can be contaminated by direct contact or, more importantly, by inhalation of off-

gassing vapors. Rescuers should avoid skin contact with these solvents or respiratory exposure in a poorly ventilated area. Even decontaminated victims exhaling these products could produce transient minor symptoms in transporting personnel. Wash oily, contaminated areas with soap and/or shampoo, if possible.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION

PETROLEUM DISTILLATES AND HALOGENATED HYDROCARBON SOLVENTS

FORMS: This group includes a wide variety of commonly used liquid hydrocarbon mixtures, many of which are highly flammable or combustible. Halogenated hydrocarbons, when exposed to heat or fire, can break down into irritant gases, including hydrochloric acid, hydrofluoric acid, or phosgene.

- Evaluate Airway *
- Oxygen – High Flow/NRM
- Irrigate injured eyes
- Cardiac Monitor **
- Transport

BASE:
Consider: IV tko

Ingestion: DO NOT induce vomiting.

NOTE: Solvent exposure alone is rarely the cause of loss of consciousness, except in cases of cardiac arrhythmia or in cases of overwhelming exposure in a confined space.

* Intubation should be considered if the victim develops severe respiratory distress.
** Arrhythmias may be delayed for up to 12 to 24 hours after exposure to Halogenated Hydrocarbon solvents. Avoid administration of EPINEPHRINE or BRONCHODILATORS.

PESTICIDES

CARBAMATES AND ORGANOPHOSPHATES

NOTES ON PESTICIDES – CARBAMATES AND ORGANOPHOSPHATES

BACKGROUND

Carbamate and organophosphate pesticides are widely used in home gardening and commercial agriculture. A variety of products are available, with widely varying potencies. They inhibit the enzyme cholinesterase, resulting in buildup of excessive acetylcholine. Unlike organophosphates, the inhibition of cholinesterase carbamates is transient and self-limited. Symptoms and signs of exposure to organophosphates or carbamates include hypersalivation, sweating, bronchospasm, abdominal cramps, diarrhea, muscle weakness, small pupils, twitching and seizures. Death is due to respiratory muscle paralysis. Nonspecific symptoms such as upper airway irritation, dizziness, nausea and headache after inhalation exposure may be due to the solvent vehicle (e.g., xylene) and not due to cholinesterase inhibition. Potential toxicity of the solvent vehicle should always be considered (see protocol for petroleum distillates).

INITIAL DECONTAMINATION PROR TO PREHOSPITAL MANAGEMENT:

Decontamination should include flushing the victim with water spray, clothing should be removed and double-bagged and skin flushed for 1 – 2 minutes. Injured eyes should be irrigated.

POTENTIAL FOR SECONDARY CONTAMINATION:

Many carbamates and organophosphates are well-absorbed through intact skin, and thus may pose a serious hazard to rescuers or health care personnel. Simple water washing may be insufficient to remove oily compounds. Wash only contaminated areas with soap and/or shampoo, if possible. Decontaminate victim thoroughly before handling them without protective clothing.

PREHOSPITAL MANAGEMENT AFTER INITIAL DECONTAMINATION:

PESTICIDES – CARBAMATES AND ORGANOPHOSPHATES

FORMS: Liquid (usually in solution with xylene or other organic solvent), solid (wettable powder). May be inhaled in an aerosol form or as a component of smoke.

- Evaluate Airway *
- Oxygen – High Flow/NRM
- Irrigate injured eyes
- Cardiac Monitor **
- Transport

BASE:
Consider: For symptomatic victims:
Atropine 0.5 – 1.0 mg I.V.;
REPEAT 2 – 4 mg. q 3 – 10 min. as needed for severe poisoning.

For Seizures: See Protocol

Ingestion: DO NOT induce vomiting.

* Intubation should be considered if the victim develops severe respiratory distress.

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POLICY MEMORANDUM 6205

DATE: 6/30/06

REVIEWED/APPROVED BY:

RICHARD C. LOTSCH, D.O., EMS AGENCY MEDICAL DIRECTOR

MICHAEL A. FRENN, EMS AGENCY ADMINISTRATOR

SUBJECT: NERVE AGENT AUTO INJECTOR POLICY

AUTHORITY: DIVISION 2.5, CALIFORNIA HEALTH & SAFETY CODE, §1797.220 & 1797.221

I. PURPOSE/POLICY:

This policy outlines the indications and procedures for the use of the Mark-1 auto injectors. This will enable those first responders trained in the use of nerve agent antidote to be able to self-administer the drug in the event of a nerve agent exposure.

II. PROCEDURE

A. The first responders must have a reasonable suspicion that they have been exposed to a nerve agent. The signs and symptoms of exposure are:

1. Unexplained runny nose
2. Tightness in the chest
3. Difficulty in breathing
4. Bronchospasm
5. Pinpoint pupils resulting in blurred vision
6. Drooling
7. Excessive sweating
8. Nausea and/or vomiting
9. Abdominal cramping
10. Involuntary urination and/or defecation
11. Jerking, twitching and staggering
12. Headache
13. Drowsiness
14. Coma
15. Convulsions
16. Apnea

<u>**Nerve Agent Mnemonic**</u>

S - Salivation
L – Lacrimation
U – Urination
D – Defecation
G - Gastrointestinal pain & gas
E - Emesis

B. If the signs and symptoms of exposure are present in the first responders, then they must use the Mark 1 kit to dose themselves.

C. Dosing is based on severity of symptoms

Signs	Symptoms	Onset	Number of auto-injector(s) to use
Vapor (small exposure)	• Pinpoint pupils • Runny nose • Mild SOB	Seconds after exposure	Mark 1 auto-injector kit (containing atropine & 2PAM): 1 dose initially; may repeat x1 in 10 minutes
Liquid (small exposure)	• Sweating • Twitching • Weak feeling	Minutes to hours after exposure	Mark 1 auto-injector kit (containing atropine & 2PAM): 1 dose initially; may repeat x1 in 10 minutes
Vapor & Liquid (large exposure)	• Convulsions • Apnea • Copious secretions	Seconds to hours after exposure	Mark 1 auto-injector kit (containing atropine & 2PAM): 3 doses initially; may repeat @ 10 minutes.

D. Training

1. Each ALS Provider Medical Director shall evaluate the training used for the Mark-1 Auto Injector, and provide documentation of training completion to the EMS Agency.
2. Training, at a minimum, shall include:
   a. Indications for self-administration
   b. Injection site location
   c. Injection dosing
   d. Arming the auto injector
   e. Administering the antidote to themselves

III. SPECIAL NOTE (SELF ADMINISTRATION)

The Solano County EMS Agency, due to regulatory constraints, does not directly authorize self-administration of medications. Any agency or provider which elects to utilize these procedures for the purpose of self-administration should obtain approval and authorization from their own risk management and/or medical director.

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EQUIPMENT

1. Mark I autoinjector antidote kit containing:
    - **Atropine** autoinjector (2 mg in 0.7 cc's)
    - **Pralidoxime chloride** autoinjector - **2-PAM** (600 mg in 2 cc's)
2. Additional atropine (2 mg) autoinjectors

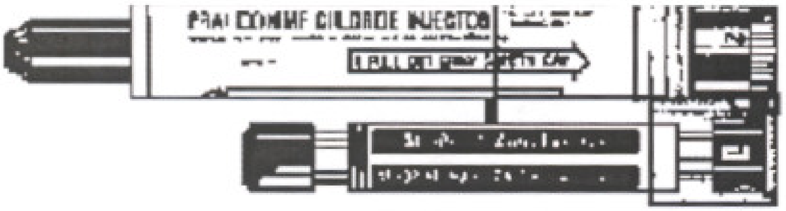


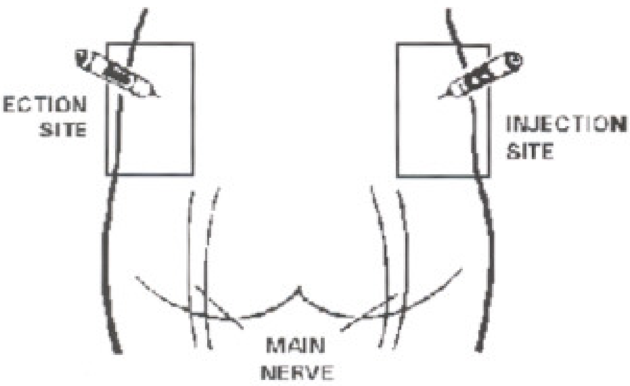
Figure 1: Mark1 Autoinjector

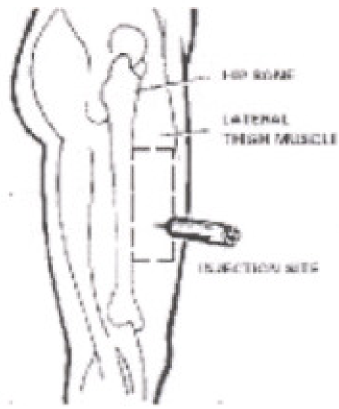
PROCEDURE

If you experience any or all of the nerve agent poisoning symptoms, you must IMMEDIATELY self-administer the nerve agent antidote.

INJECTION SITE SELECTION

- The injection site for administration is normally in the **outer thigh muscle.** It is important that the injections be given into a large muscle area.
- If the individual is thinly-built, then the injections should be administered into the
the
upper outer quadrant of the buttocks





Figures 2 and 3: Buttock and thigh sites for self-administration

ARMING THE AUTO INJECTOR

- Immediately put on your protective mask.
- Remove the antidote kit
- With your nondominant hand, hold the autoinjectors by the plastic clip so that the larger autoinjector is on top and both are positioned in front of you at eye level.
- With your dominant hand grasp the **atropine** autoinjector (the smaller of the two) with the thumb and first two fingers. DO NOT cover or hold the needle end with your hand, thumb, or fingers-**you might accidentally inject your self**. An accidental injection into the hand WILL NOT deliver an effective dose of the antidote, especially if the needle goes through the hand.

Pull the injector out of the clip with a smooth motion. The autoinjector is now armed.

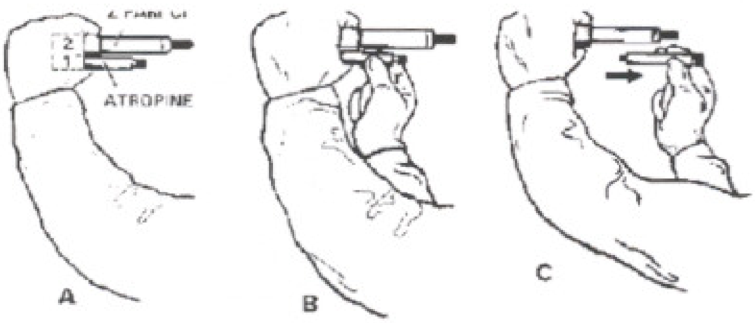


Figure 4: arming the autoinjector

ADMINISTRATION

- Hold the **Atropine** portion of the autoinjector with your thumb and two fingers (pencil writing position). Be careful not to inject yourself in the hand!
- Position the green (needle) end of the injector against the injection site (thigh or buttock). DO NOT inject into areas close to the hip, knee, or thigh bone.
- Apply firm, even pressure (not jabbing motion) to the injector until it pushes the needle into the thigh (or buttocks). Using a jabbing motion may result in an improper injection or injury to the thigh or buttocks.
- Hold the injector firmly in place for at least 10 seconds. Firm pressure automatically triggers the coiled spring mechanism. This plunges the needle into the muscle (through the clothing if necessary) and at the same time injects the antidote into the muscle tissue.

- Carefully remove the **Atropine** autoinjector from the injection site.
- Next pull the **2 PAM** injector (the larger of the two) out of the clip.
- Inject the **2 PAM** in the same manner as the steps above, holding the black (needle) end against the outer thigh (or buttocks).
- Massage the injection sites, if time permits.
- After administering the first set of injections, wait 5 to 10 minutes.
- After administering one set of injections, you should initiate decontamination procedures, as necessary, and put on any additional protective clothing.

Atropine only may be repeated every 10 - 15 minutes as needed.